

MS Electrical and Computer Engineering (Machine Learning and Data Science)

General description

The modern world is filled with digital data, arising both from sensor recordings and other information sources. Every day, the amount of data (from audio to video, and from electronic health records to browsing interests and shopping preferences) continues to increase, and there is correspondingly an explosion of demand from industry for engineers who are trained to extract relevant information from such data so that it can be used to improve modern technology and society. This MS degree will provide students with rigorous training in the theory, methods, and applications of data science, machine learning, and signal and information processing. This degree program has been designed specifically for students with a background in electrical and computer engineering or a closely related field, and who intend to use machine learning or data science as a substantial part of their career. It includes courses in pertinent aspects of software and software engineering, relevant mathematics, machine learning and optimization, deep learning, relevant signal processing knowledge (particularly useful for data that has inherent structure), and a wide variety of application areas to choose from. This degree program can be viewed as a focused version of the more general Master of Science in Electrical Engineering. For students interested specifically in machine learning and data science, this more focused degree program will provide a deeper understanding of this subarea of EE, and a more complete and focused skill set for work in this area.

Admission requirements

For newly incoming students, refer to [USC Viterbi School of Engineering](#). For students enrolled in a current USC MS program in ECE, admission requires completion of at least two of the courses EE 483, EE 503, EE 510, and a minimum GPA requirement may be enforced depending on space available and number of applicants.

Program beginning date and transition

This revised degree program starts Fall 2021 semester, and applies to all students starting their MSECE (MLDS) program in the Fall 2021 semester or later. Students who enrolled in the MS ECE (MLDS) program prior to Fall 2021, may complete their program under the degree requirements in effect when they began the (original) degree program, or may opt to change to the new degree program requirements. Students following the original program requirements will now be allowed to substitute the previous 'Software' course requirement by taking both of the two new software courses EE 538 and EE 547. Other course substitutions that are in line with the new program requirements may also be allowed and will be considered on a case-by-case basis.

